

From free field theory to symmetric polynomials

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Joint work with Akihiro Tsuchiya ([arXiv:1302.6435](https://arxiv.org/abs/1302.6435)) and ongoing work with David Ridout

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Outline

- 1 Representation theory from singular vectors
- 2 Free field theories and screening operators
- 3 Symmetric polynomials
- 4 Application to the minimal models

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Virasoro Verma modules are representations of the chiral algebra.

Def: Virasoro Verma module $M(h)$, $h \in \mathbb{C}$

- Generated by $|h\rangle$: $L_0|h\rangle = h|h\rangle$, $L_n|h\rangle = 0, n \geq 1$
- $M(h) = U(\bigoplus_{n \geq 1} \mathbb{C}L_{-n})|h\rangle$

Minimal models

Let $(p_+, p_-) = 1, p_+, p_- \geq 2$.

The minimal model chiral algebra

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Minimal model representation theory

The representations of M_{p_+, p_-} are those of $V(c_{p_+, p_-})$ on which fields corresponding to states in $I(p_+, p_-)$ act trivially.

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Goal for the rest of the talk: Find a better way to get these constraints for the example of the minimal models.

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- The minimal models can be realised as either a quotient of the Virasoro chiral algebra or a **subquotient of the Heisenberg chiral algebra**.

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The simple $\mathcal{H}$ -modules are given by Fock spaces.

Def: Fock spaces $F_\lambda, \lambda \in \mathbb{C}$

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- $L_0|\lambda\rangle = \frac{1}{2}(a_0^2 - \alpha_0 a_0)|\lambda\rangle = h_\lambda|\lambda\rangle$
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- $V_\lambda(z) : F_\mu \rightarrow F_{\lambda+\mu}[[z, z^{-1}]] z^{\lambda\mu}$

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- The map $S_0 : F_\mu \rightarrow F_{\mu+\lambda}$ is only well defined if $\lambda\mu \in \mathbb{Z}$.
- If $\lambda\mu \in \mathbb{Z}$ and $S_0|\mu\rangle \neq 0$ then $S_0|\mu\rangle$ is a **singular vector** in $F_{\mu+\lambda}$

Composing screening operators

If $\lambda\mu$ is not an integer, one must consider higher powers of the screening operator. Recall $\lambda = \alpha_{\pm}$, $\alpha_+ = \sqrt{\frac{2p_-}{p_+}}$, $\alpha_- = -\sqrt{\frac{2p_+}{p_-}}$.

$\lambda = \alpha_+$ **case:**

$$S_+(z_1) \cdots S_+(z_r) =$$

$$e^{r\alpha_+ \hat{a}} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{\alpha_+ a_0} \prod_{m \geq 1} e^{\alpha_+ \frac{a_{-m}}{m} \sum_i z_i^m} \prod_{m \geq 1} e^{-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}}$$

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Thm: [Tsuchiya, Kanie], [Varchenko]

If and only if $\mu = \frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+}$, $k \in \mathbb{Z}$ then there exists a unique choice of a cycle Γ such that

$$S_+^{[r]} = \int_{\Gamma} S_+(z_1) \cdots S_+(z_r) dz_1 \cdots dz_r : F_{\frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+}} \rightarrow F_{\frac{1+r}{2}\alpha_+ + \frac{k}{\alpha_+}}$$

is a non-trivial Virasoro homomorphism.

- if $k \geq 0$ then $S_+^{[r]} | \frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+} \rangle = 0$.
- if $k < 0$ then $S_+^{[r]} | \frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+} \rangle \neq 0$.

The singular vector in the Heisenberg chiral algebra

The singular vector that generates the ideal by which one must quotient the Heisenberg chiral algebra to get the minimal model chiral algebra is just

$$S_+^{[p_+-1]}|(1-p_+)\alpha_+\rangle \in \mathcal{H}$$

or

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The constraints on the minimal model conformal weights are obtained by evaluating the action of the zero mode $(S_+^{[p_+-1]}|(1-p_+)\alpha_+\rangle)_0$ or $(S_-^{[p_--1]}|(1-p_-)\alpha_-\rangle)_0$.

Appearance of symmetric polynomials

Action of $S_+^{[r]}$ on $F_{\frac{1-r}{2}\alpha_+ + \frac{k}{\alpha_+}}$:

$$S_+^{[r]} = e^{r\alpha_+ \hat{a}} \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{(1-r)\frac{\alpha_+^2}{2}} \prod_{i=1}^r z_i^{k+1} \\ \times \prod_{m \geq 1} e^{\alpha_+ \frac{a-m}{m} \sum_i z_i^m} \prod_{m \geq 1} e^{-\alpha_+ \frac{a_m}{m} \sum_i z_i^{-m}} \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

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This integral defines a well known inner product on the ring of symmetric polynomials $\mathbb{C}[z_1, \dots, z_r]^{\mathfrak{S}_r}$:

$$\langle f, g \rangle_r^{\alpha_+^2/2} = \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{(1-r)\frac{\alpha_+^2}{2}} f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

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$$\langle f, g \rangle_r^{\alpha_+^2/2} = \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^r z_i^{(1-r)\frac{\alpha_+^2}{2}} f(z_1^{-1}, \dots, z_r^{-1}) g(z_1, \dots, z_r) \frac{dz_1 \cdots dz_r}{z_1 \cdots z_r}$$

This inner product is symmetric and it vanishes if $\deg(f) \neq \deg(g)$.

Outline

- 1 Representation theory from singular vectors
- 2 Free field theories and screening operators
- 3 Symmetric polynomials**
- 4 Application to the minimal models

Symmetric polynomials

- One can consider symmetric polynomials in either a finite or infinite number of variables:
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- There is a well known basis that is orthogonal with respect to the inner product $\langle f, g \rangle_r^{\alpha_+^2/2}$ called Jack symmetric polynomials.

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- The inner product of a Jack polynomial $J_\nu(z; \frac{\alpha_+^2}{2})$ with itself

$$\langle J_\nu(z; \frac{\alpha_+^2}{2}), J_\nu(z; \frac{\alpha_+^2}{2}) \rangle_r^{\alpha_+^2/2}$$

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- $$\prod_{m \geq 1} e^{\frac{\alpha_{\pm}^2}{2} \frac{p_m(y)p_m(z)}{m}} = \sum_{\nu} b_{\nu}(\frac{\alpha_{\pm}^2}{2}) J_{\nu}(y; \frac{\alpha_{\pm}^2}{2}) J_{\nu}(z; \frac{\alpha_{\pm}^2}{2})$$

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Explicit singular vector formulae [Mimachi, Yamada]

Let

$$\begin{aligned}\rho : \mathbb{C}[p_1, p_2, \dots] &\rightarrow \mathbb{C}[a_{-1}, a_{-2}, \dots] \\ p_k(y) &\mapsto \frac{2}{\alpha_+} a_{-k} .\end{aligned}$$

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$$\begin{aligned}
 & S_+^{[p_+-1]} |(1-p_+)\alpha_+\rangle \\
 &= \int_{\Gamma} \prod_{i \neq j} (z_i - z_j)^{\alpha_+^2/2} \prod_{i=1}^{p_+-1} z_i^{(2-p_+)\frac{\alpha_+^2}{2}} \prod_{i=1}^{p_+-1} z_i^{1-p_-} \prod_{m \geq 1} e^{\alpha_+ \frac{a-m}{m} p_m(z)} |0\rangle \frac{dz_1 \cdots dz_{p_+-1}}{z_1 \cdots z_{p_+-1}} \\
 &= \langle \prod_{i=1}^{p_+-1} z_i^{p_- - 1}, \rho(\prod_{m \geq 1} e^{\frac{\alpha_+^2}{2} \frac{p_m(y)p_m(z)}{m}}) \rangle_{p_+-1}^{\alpha_+^2/2} |0\rangle \\
 &= \sum_{\nu} \langle \prod_{i=1}^{p_+-1} z_i^{p_- - 1}, J_{\nu}(z; \frac{\alpha_+^2}{2}) \rangle_{p_+-1}^{\alpha_+^2/2} b_{\nu}(\frac{\alpha_+^2}{2}) \rho(J_{\nu}(y; \frac{\alpha_+^2}{2})) |0\rangle
 \end{aligned}$$

This proof: [Tsuchiya, SW]

$$\prod_{i=1}^{p_+-1} z_i^{p_- - 1} = J_{(p_- - 1)p_+ - 1}(z; \frac{\alpha_+^2}{2})$$

$$S_+^{[p_+-1]} |(1-p_+)\alpha_+\rangle = \rho(J_{(p_- - 1)p_+ - 1}(y; \frac{\alpha_+^2}{2})) |0\rangle$$

Getting the conformal weights from the zero mode

The zero mode whose action we need to evaluate is the zero mode of the field

$$\begin{aligned} & S_+^{[p_+-1]} V_{(1-p_+)\alpha_+}(w) \\ &= \int_{\Gamma} S_+(z_1 + w) \cdots S_+(z_{p_+-1} + w) V_{(1-p_+)\alpha_+}(w) dz_1 \cdots dz_{p_+-1}. \end{aligned}$$

The easiest way to evaluate such a zero mode is to evaluate matrix elements

$$\langle \tau | S_+^{[p_+-1]} V_{(1-p_+)\alpha_+}(w) | \tau \rangle,$$

where $\tau \in \mathbb{C}$ and $\langle \tau | a_{-n} = 0, n \geq 1$.

Getting the conformal weights from the zero mode

$$\begin{aligned}\langle \tau | S_+^{[p_+-1]} V_{(1-p_+)\alpha_+}(w) | \tau \rangle &= \dots \\ &= \langle J_{(p_--1)p_+-1}(z; \frac{\alpha_+^2}{2}), \prod_{i=1}^{p_+-1} (1 + \frac{z_i}{w})^{\alpha_+ \tau} \rangle_{p_+-1}^{\alpha_+^2/2}\end{aligned}$$

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There are known formulae for expanding

$$\prod_{i=1}^{p_+-1} (1 + \frac{z_i}{w})^{\alpha_+\tau} = \sum_{\nu} g_{\nu}(\alpha_+\tau) J_{\nu}(z; \frac{\alpha_+^2}{2}) w^{-|\nu|}$$

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There are known formulae for expanding

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$$\langle \tau | S_+^{[p_+-1]} V_{(1-p_+)\alpha_+}(w) | \tau \rangle \propto \prod_{r,s} (h_{\tau} - h_{r,s})$$

$$h_{r,s} = h_{p_+ - r, p_- - s} = \frac{(p_+ s - p_- r)^2 - (p_+ - p_-)^2}{4p_+ p_-}$$

Applicability of these free field methods

- Today's result: Symmetric polynomials can be used to “quite easily” classify the irreducible minimal model representations.
- Essentially the same method also works for free field realisations of affine $\mathfrak{sl}_2$. [Ridout, SW; coming soon]
- This method was a crucial tool in the classification of the simple modules of certain logarithmic extensions of the minimal models. [Tsuchiya, SW]

The End

Thank you for your attention!